VASANTHKUMAR N

Mechanical Engineering | B.E. (2023–2027) | Kamaraj College of Engineering & Technology
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OBJECTIVE

Final year B.E. Mechanical Engineering student (Class of 2027) with a strong foundation in CAD modelling (Fusion 360, CATIA V5, Creo), FEA simulation (ANSYS), and IoT-based systems. Seeking an entry-level Design Engineering or Manufacturing Engineer role to apply hands-on fabrication and simulation skills to innovative product development.

EDUCATION

Institution	Course	Percentage / CGPA	Year
Kamaraj College of Engineering & Technology	B.E. Mechanical Engineering	CGPA: 7.85/10 (up to Semester 6)	2023–2027
Amudham Matric Hr. Sec. School	HSC	71%	2023

TECHNICAL SKILLS

- CAD Software: AutoCAD (2D Drafting), PTC Creo (3D Parametric CAD), Autodesk Fusion 360, CATIA V5
- Simulation / FEA: ANSYS Workbench (Static Structural, Steady-State Thermal, Modal Analysis)
- IIoT & Automation: ESP8266 Wi-Fi, RFID Telemetry, Sensors (ADXL345 Vibration, DS18B20 Temp), Firebase
- Programming: Python (Machine Learning — Random Forest, data pipelines)
- Manufacturing & QC: CNC Programming (G-Code/M-Code), GD&T Tolerancing, Metrology & Inspection
- 3D & Media Tools: Blender (3D Modeling & Rendering), DaVinci Resolve (Video Production)

PROJECTS

Digital Twin-Based Predictive Maintenance System — *Capstone Project, 2026–27*

- Developing a Digital Twin framework integrating ANSYS FEA, ESP8266 IoT telemetry, Firebase cloud, Unity 3D, and Machine Learning for real-time health monitoring of rotating machinery.
- Engineering a real-time data acquisition system with ADXL345 3-axis vibration and DS18B20 temperature sensors mounted on motor bearing housings.
- Implementing a Random Forest ML model to detect and classify bearing faults, dynamic imbalance, and shaft misalignment for early failure prevention.
- Designing a synchronized 3D Unity digital twin dashboard displaying live sensor data to deliver an affordable predictive maintenance solution for MSMEs.

Design & Fabrication of a Compact Heat Recovery System — *3rd Year Project, 2024–25*

- Designed a shell-and-helical-coil heat exchanger for industrial waste heat recovery in Autodesk Fusion under compact envelope constraints.
- Conducted thermal and static structural stress simulation in ANSYS Workbench to validate fluid velocity, heat transfer coefficients, and shell pressure drops.
- Fabricated and tested a physical working prototype; measured heat transfer effectiveness and fluid flow parameters across varying operating conditions.
- Optimized helical coil pitch geometry and shell dimensions to achieve high thermal effectiveness under compact form-factor constraints.

Smart Non-Newtonian Fluid Dynamic Speed Bump — *Thinkathon Innovation, KCET*

- Engineered a concept for an adaptive speed bump utilizing shear-thickening non-Newtonian fluid (STF) that yields under low-speed vehicle roll while rigidifying into a solid obstacle upon high-speed impact.
- Conceived a passive mechanism enabling emergency vehicles (ambulances, fire engines) to traverse bumps smoothly without jarring chassis deceleration.
- Designed a reinforced elastomeric chamber housing layout engineered to resist cyclic heavy axle loads and prevent fluid leakage.

INTERNSHIPS & INDUSTRIAL EXPERIENCE

Pentagon Apparels Unit II, Virudhunagar — *IoT / RFID Sewing Performance Tracking System — [Add Dates]*

- Implemented an RFID-based shop-floor tracking system using ESP8266 Wi-Fi modules to monitor operator-level sewing output in real time across manufacturing lines.
- Integrated Wi-Fi telemetry with a live centralized web dashboard, enabling plant supervisors to identify production bottlenecks and balance line throughput instantly.
- Eliminated manual paper logging errors and reduced data reconciliation latency across active production shifts.

Paint Shop Department Internship — *15 Days*

- Observed end-to-end industrial finishing operations including chemical surface pre-treatment, primer/topcoat spray application, and oven curing stages.
- Studied quality control checkpoints for finish consistency, coating thickness (DFT), and surface defect diagnosis.
- Gained practical exposure to industrial safety protocols, personal protective equipment (PPE), and workflow sequencing in an active manufacturing plant.

ACHIEVEMENTS, HONORS & MEMBERSHIPS

- 2nd Prize — Technical Quiz, Yuvasakthi 2K26 National Level Technical Symposium, NPR College of Engineering & Technology, Natham (Mar 2026)
- 3rd Prize — SkyForge Event, Euphoria Techno Management Meet, Kalasalingam Academy of Research & Education (Mar 2026)
- ISTE Student Member (SM No. 11709), Indian Society for Technical Education (2025–2027)

CERTIFICATIONS & PROFESSIONAL TRAINING

- NPTEL Elite Certificate — Product Design & Manufacturing, IIT Kanpur / SWAYAM (Jan–Apr 2026)
- Industrial Internet of Things (IIoT) Hands-on Training, SASTRA University, Thanjavur
- CATIA V5 Certification, Kamaraj College of Engineering & Technology
- Advanced Creo (3D CAD) Professional Certificate, Udemy
- CNC Programming Training (G-Code/M-Code), NTTF (Nettur Technical Training Foundation), Bangalore

SOFT SKILLS

Problem-Solving • Creative Thinking • Cross-Functional Team Collaboration

TECHNICAL INTERESTS

3D Parametric Part Modelling • CAD/CAM Automation • Industrial IoT Telemetry

DECLARATION

I hereby solemnly declare that all the information furnished above is true, complete, and correct to the best of my knowledge and belief.